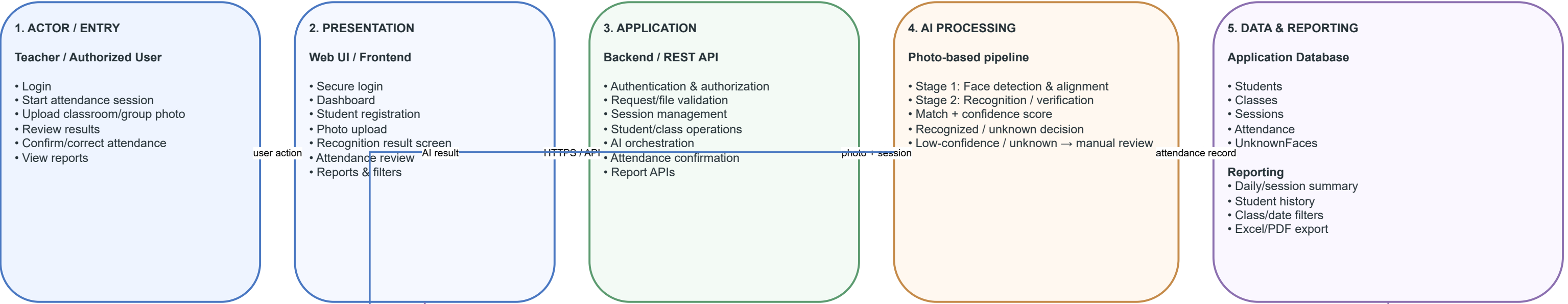
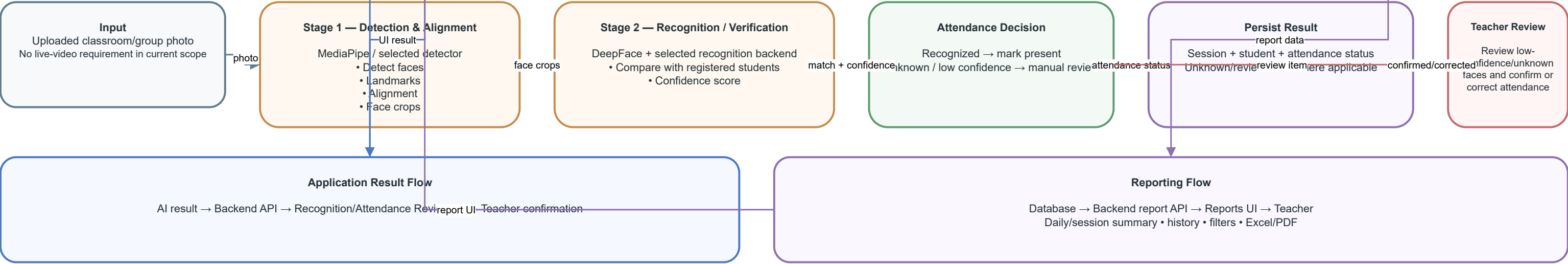


AI Smart Attendance System — High-Level Architecture (Day 2)

Uploaded classroom/group photo workflow • Detection → Recognition → Attendance → Reporting



AI Decision Pipeline — Detailed View



SECURITY & DATA PROTECTION — CROSS-CUTTING

- Authentication & authorization for protected operations
- Validate uploaded files and request fields
- Keep passwords, real student photos, raw embeddings, and sensitive reports out of Git
- Store sensitive artifacts only in approved private/local storage
- Restrict access to attendance and reports

Technology note: Backend can use Flask/FastAPI; frontend React; development DB SQLite with PostgreSQL as production candidate; AI stack includes MediaPipe/OpenCV and DeepFace. Final model/configuration is subject to empirical testing on the project dataset.

Scope: Day-2 high-level architecture • Current attendance input is an uploaded classroom/group photo • No live-video attendance in current scope